

LUMINA ENERGY · HELIOS PLATFORM

Release Notes

Versions v4.4.0 – v4.7.1 · Last 8 Releases

Maintained by: Priya Nair (Release Management) & Engineering team | Cadence: every 6–8 weeks | Generated: 2024-11-10

v4.7.1 ← CURRENT	2024-11-05	PATCH
Release Manager: Priya Nair	Deploy duration: 14 min (no DB migration)	

- BUG FIXES
- [Grid Intelligence] Outage confidence score returned 0 for substations with >500 meters — uint16 overflow in weighted average. Affected NW Grid UK, ENGIE, TransnetBW since v4.7.0. — Farah Okonkwo
 - [Portal] Timezone-aware date picker ignored DST transitions for CET customers (German, French). — Clara Dupont
 - [Notifications] Duplicate outage-restoration notifications sent when outage resolved within 30s of initial notification (race condition). — Yuki Nakamura
 - [API] GET /v3/meters/:meterId/readings returned HTTP 500 instead of 404 for cross-tenant meter IDs (minor security hardening). — David Okafor / Yasmin Osei

- INFRASTRUCTURE
- EKS upgraded to Kubernetes 1.30 (from 1.29). — Rosa Lindqvist
 - IoT device CA intermediate certificate rotated (scheduled 90-day cycle). — Lars Eriksson

v4.7.0	2024-10-14	MAJOR
Release Manager: Tanvir Rahman	Deploy duration: 28 min (DB migration + cache warm-up)	

- NEW FEATURES
- HELIOS-1887 · Outage Confidence Scoring — Each detected outage now carries a 0–100% confidence score. High-confidence (≥85%) auto-escalates to dispatch; low-confidence (40–84%) surfaces for manual review. Estimated 30% reduction in false-positive dispatches. — Farah Okonkwo, Chidi Eze
 - HELIOS-1856 · Batch Device Configuration API — POST /v3/devices/config/batch pushes config to up to 1,000 devices in a single call. — Lars Eriksson
 - HELIOS-1799 · Analytics Data Export v2 — Asynchronous exports (202 Accepted + job ID), XLSX/CSV, date ranges up to 3 years. — Lin Chen, Ravi Krishnan

- IMPROVEMENTS
- Grid telemetry heatmap refresh: 60s → 30s. — Chidi Eze
 - Dispatch mobile job history extended: 30 → 90 days. — Ami Tanaka
 - API auth errors return structured JSON with error_code. — David Okafor
 - X-Request-ID header added to all API responses. — Priya Nair
 - TimescaleDB continuous aggregate midnight-boundary chunk skip fixed; affected data backfilled. — Rosa Lindqvist

- GIS map auto-paginates GeoJSON for customers with >20,000 substations. — Chidi Eze

BREAKING CHANGES

- GET /v3/analytics/export now returns 202 Accepted — clients must poll GET /v3/analytics/export/{job_id}. Affected clients notified 30 days in advance; migration guide distributed.

v4.6.1	2024-10-08	CRITICAL PATCH
Release Manager: Priya Nair	Deploy duration: Critical fixes for NW Grid UK pilot	

BUG FIXES

- [Mobile] SQLite migration crash on Samsung A34/A54 on first launch (SQLite < 3.37). Added compatibility handling. — Ami Tanaka
- [Mobile] Photo sync uncaught exception when >80 photos queued. Fixed with 20-photo batch ceiling. — Ami Tanaka
- [Mobile] Offline job updates not visible in portal until manual sync. Now auto-syncs on reconnect. — Ami Tanaka
- [Dispatch API] Returns 409 (not 500) for already-assigned job assignment attempt. — Farah Okonkwo

v4.6.0	2024-09-23	MAJOR
Release Manager: Tanvir Rahman	Deploy duration: 32 min (schema changes + ArgoCD sync)	

NEW FEATURES

- HELIOS-1891 · Technician Mobile Offline Mode — View jobs, update status, add notes, attach photos without connectivity. Auto-syncs on reconnect with conflict resolution. Piloting with NW Grid UK (40 technicians). GA: Q4 2024. — Ami Tanaka
- HELIOS-1832 · Device Firmware OTA Scheduling — Schedule fleet firmware updates with target group, maintenance window, and auto-rollback criteria. — Lars Eriksson
- HELIOS-1788 · Grid Health Score Dashboard — Composite 0–100 score per customer network (device connectivity, outage frequency, forecast accuracy, telemetry freshness). — Chidi Eze

IMPROVEMENTS

- Kafka consumer group rebalancing: ~45s → ~15s via cooperative rebalancing protocol. — David Okafor
- Auth token expiry returns error_code: TOKEN_EXPIRED. — Yasmin Osei
- Customer portal initial load improved ~22% via lazy-loading grid topology. — Mateus Costa

SECURITY

- Patched express dependency (CVE-2024-XXXX — path traversal in static serving; not exploitable in Helios config). — Yasmin Osei

v4.5.0 – v4.5.2	2024-08-12 – 2024-09-02	MAJOR + PATCHES
Release Manager: Priya Nair	Deploy duration: 45 min — largest release of year (multiple Kafka topics + DB migrations)	

NEW FEATURES (V4.5.0)

- HELIOS-1820 · Demand Forecast Accuracy Dashboard — 7/30/90-day rolling MAE, MAPE, RMSE. Forecast vs. actual chart. SLA breach highlighting. Piloted with ENGIE and NW Grid UK. — Lin Chen
- HELIOS-1750 · Multi-Region Australian Support — Ausgrid and AusNet data stored and processed in ap-southeast-2 (Sydney) for data residency compliance. — Rosa Lindqvist, Priya Nair
- HELIOS-1776 · Outage RCA Categories — Technicians assign root cause on job close; feeds OFGEM/BDEW regulatory reports. — Farah Okonkwo

INFRASTRUCTURE (V4.5.0)

- TimescaleDB upgraded 2.11 → 2.14 (~15% storage reduction). — Rosa Lindqvist
- PostgreSQL (Aurora) upgraded 14 → 15. — Rosa Lindqvist

PATCH FIXES (V4.5.1 / V4.5.2)

- v4.5.1: Fixed MAPE divide-by-zero when actual demand is zero. — Lin Chen
- v4.5.2: Forecast SLA breach threshold now read from customer config (not hardcoded). Affected ENGIE + NW Grid UK. — Mateus Costa

v4.4.0	2024-05-28	MAJOR
Release Manager: Tanvir Rahman	Deploy duration: —	

NEW FEATURES

- HELIOS-1654 · Automated Outage Severity Escalation — Severity re-evaluated every 5 min while active; auto-upgrades dispatch priority and notifies stakeholders. — Farah Okonkwo
- HELIOS-1698 · Bulk Meter Registration API — POST /v3/meters/bulk registers up to 10,000 meters in one call. Returns job ID. — Lars Eriksson
- HELIOS-1631 · Customer API Usage Dashboard — Operators can view API usage metrics (calls/day, top endpoints, error rates). — Priya Nair

BUG FIXES

- Outage notifications now sent to all contacts (not only contacts[0]). Regression from v4.3.0; see INC-2024-0312. — Yuki Nakamura

Older release notes (v1.0–v4.3) are in [Confluence: Engineering > Release History](#). For upcoming releases see the [Product Roadmap](#) and [Current Sprint Goals](#) documents.